CASE FILE NO. 2022-262 — OPENED FOR REVIEW

DRAFTSPLINE

Every NFL draft pick since 2012, graded against what its slot actually produces — not what a decades-old chart says it should.

24.48×

[OUTLIER — UNADJUSTED]

Brock Purdy — pick #262, 2022 — produced 24.48× his slot's historical expected value. The largest True Draft Value Score in the dataset, bigger than any first-round pick ever recorded.

Submission document · full project writeup, methodology, and technical reference

Project Submission

Project name

DraftSpline

Elevator pitch

DraftSpline grades every NFL draft pick since 2012 against what its slot actually produced in real on-field value, not what a decades-old draft chart says it should be worth, then uses that same model to project the 2026 class and lets you ask an AI analyst to explain any pick, team, or trend in plain English.

Tagline options

- The draft chart gets a performance review.
- Stop trusting a chart from 1960. Trust the tape.
- Every pick, graded on what actually happened.
- Draft value, recalculated from real production.
- The scouting report your gut feeling never had.

Inspiration

Every draft analyst leans on the same traditional draft value chart, a static table built decades ago that says pick 1 is worth roughly 33 times as much as pick 256. It has never been updated with real outcomes. We wanted to know what happens if you throw that out and instead ask the data directly: given everything we know about every drafted quarterback, running back, receiver, and tight end since 2012, what does a given pick actually tend to produce? That question became TDVS, our True Draft Value Score, and the rest of the app grew around it.

What it does

DraftSpline pulls play-by-play and draft data from nflverse, computes rookie-contract-era EPA (Expected Points Added) for every modeled position, and fits a data-driven expected-value curve per draft slot. Every pick gets a TDVS score: actual production divided by what that slot has historically produced. From there:

- The Draft Board shows every class from 2012 to 2025 with TDVS, bust rates, and searchable filters.
- The Curve Comparison page overlays our EPA curve against the traditional chart with a labeled confidence band, so you can see exactly where conventional wisdom and real outcomes diverge.
- The Redraft Simulator reorders any complete draft class by actual value produced, with team-by-team deltas.
- The GM Scorecard ranks all 32 franchises by drafting efficiency.
- The 2026 Draft Preview joins this year's incoming class to the same curve to project expected value before a single rookie snap is played.

- The AI Analyst answers natural-language questions about steals, busts, and team trends, grounded entirely in our own dataset so it cannot fabricate players or stats.

How we built it

The backend is a Python and FastAPI service backed by a five-stage data pipeline: pull raw nflverse data, compute EPA per player per season, calculate TDVS scores with statistical stabilization for near-zero denominators, fit the expected-value curve using kernel smoothing and isotonic regression, then roll everything up into team-level scores. Results are written to parquet files and served from an in-memory store. The frontend is React with Vite, using Recharts for all data visualization, with a custom design system styled around a scouting-room aesthetic rather than a generic dashboard look. The AI Analyst is powered by Groq's LLM API, with every response grounded in a JSON context block built from real computed statistics rather than free generation.

Challenges we ran into

Running back EPA baselines sit close to zero, which made TDVS blow up toward infinity for some players when the denominator approached zero. We fixed this with a continuous stabilization formula instead of an arbitrary clamp. Our first curve-fitting attempt using a spline produced wildly unrealistic values at certain picks, which we replaced with kernel smoothing plus isotonic regression to guarantee a sane, monotonic curve. We also had to be careful that incomplete recent draft classes, like 2023 through 2025, did not get judged the same way as classes with a full four-year rookie window, since that was both statistically misleading and an easy way for the AI Analyst to say something indefensible.

Accomplishments that we're proud of

The single number that anchors the whole project, Brock Purdy's TDVS of 24.48 as a pick-262 quarterback, is both the most attention-grabbing stat in the app and the one we worked hardest to caveat honestly on screen, rather than just bury in a methodology page. We are also proud that the AI Analyst is fully grounded: it cannot invent players or numbers, because every claim it makes has to trace back to a real computed statistic in our dataset.

What we learned

We learned how much nuance lives inside a metric that looks simple from the outside. Actual versus expected sounds straightforward until you have to decide what counts as a qualifying season, how to handle a position whose baseline production hovers near zero, and how to talk about an outlier result without either hiding it or overselling it.

What's next for DraftSpline

Extending TDVS to defensive positions and offensive line, adding trade-chart data so the 2026 preview can reflect actual draft-day trades instead of just original slot value, and building year-over-year tracking so a front office could watch its drafting efficiency trend over multiple seasons.

Built with

Python, FastAPI, pandas, pyarrow, scikit-learn, nfl_data_py, Groq API, React, Vite, React Router, Recharts, JavaScript, HTML, CSS, Render, Vercel, GitHub.

The Finding

Using EPA from nflverse across 14 draft classes (2012-2025), DraftSpline finds:

- The 7th-round pick #262 in 2022 — Brock Purdy, "Mr. Irrelevant" — produced 24.48× his slot's expected value. That's the single largest TDVS in the dataset, larger than every first-round pick in any class modeled. We lead with this number and immediately flag why it's contestable: TDVS doesn't adjust for offensive line quality, receiving talent, or scheme, and Purdy's score is in part a stress-test of that blind spot.
- Quarterback has the highest bust rate of any modeled position: 55.2% of qualifying QB picks miss their slot's expectation (TDVS < 0.5), vs. 43.1% for RB, 39.9% for WR, and 35.6% for TE — measured on the 2012-2022 subset where every player has a complete 4-year rookie window, so the comparison is apples-to-apples.
- The New York Jets have drafted −1,096.9 EPA below expectation since 2012 — the worst accumulated drafting record of any of the 32 franchises, while the 49ers lead at +301.8.
- 2026 Draft Preview: Carnell Tate (WR, pick #4 to Tennessee) is the highest-ceiling pick by expected slot value (76.1 EPA); quarterback is the riskiest rounds 1-2 bet at a 55.2% historical bust rate. Trade-up cost isn't shown — nflverse has no original-team-per-pick field, so it's reported as unavailable rather than guessed.

Every pick is scored by TDVS (True Draft Value Score): the EPA a player actually produced during their 4-year rookie contract, divided by what was historically expected from that draft slot. TDVS > 1.0 means a player outperformed his slot; the formula reframes the draft as a capital allocation problem instead of a talent-guessing game.

Features

Landing page (/) — A single case-file-style page built around the project's boldest claim: Purdy's 24.48x TDVS, shown with the same outlier caveat used everywhere else in the app, plus three supporting headline stats and one call-to-action into the Draft Board.

TDVS Draft Board (/board) — Every pick in a selected class (2012-2025), toggleable between pick order and TDVS ranking with an animated reorder. Defaults to hiding unmodeled positions with a toggle to reveal them; includes a player-name search box and position filter tabs. Shows three hero stats and the cross-position bust-rate chart on load.

Player drill-down — Click any pick to open a modal with year-by-year EPA bars, an auto-generated one-sentence verdict, a limitation caveat on any outlier score (TDVS ≥ 5), and a deep link into the Redraft Simulator for that class.

Draft Value Curve Comparison (/curve) — The traditional Stuart Chase chart vs. DraftSpline's per-position EPA curve, both normalized to pick #1 = 1.0. The 90% confidence band is labeled in the legend, and two dynamic callouts state the exact pick-number divergence and CI-width growth.

Redraft Simulator (/redraft) — Re-orders a class by TDVS and shows original vs. optimized order with hover-linked rows and a per-team value-delta bar chart. Restricted to 2012-2022 classes (complete rookie windows only).

AI Draft Analyst (/analyst) — Groq-backed chat grounded in the real TDVS data. Every request includes cross-position bust/steal rates; an explicit guard prevents the model from inventing players, scores, or seasons outside the provided context.

GM Scorecard (/teams) — All 32 franchises ranked by capital-weighted TDVS, with best/worst-franchise hero cards at the top.

Methodology page (/methodology) — Plain-language writeup of TDVS, the curve-fitting method, and every known limitation.

2026 Draft Preview (/preview) — Every 2026 pick joined to the same expected-EPA curve used everywhere else in the app, with hero stats and a per-team "expected EPA acquired" bar chart.

How It Works

Data source: nflverse play-by-play (2012-2025) and draft pick data, pulled via nfl_data_py. Classes 2023-2025 are scored as preliminary (rookie windows incomplete); only 2012-2022 classes feed curve fitting and the Redraft Simulator.

TDVS formula: rookie_epa_total / expected_epa[pick][position], computed across Years 1-4 of each player's rookie contract, with games-played normalization for injury-shortened seasons. Stabilized near zero/negative expected_epa with a formula mathematically identical to the literal ratio whenever expected_epa is comfortably positive.

Curve fitting: The expected-EPA curve is fit per position (QB/RB/WR/TE) using Nadaraya-Watson kernel smoothing followed by isotonic regression, with a 90% confidence band that widens in sparse late rounds.

Simulation logic: The Redraft Simulator re-orders all qualifying players in a complete-window class by TDVS descending and reassigns them to the original pick slots, then computes each team's value delta against what they actually drafted.

Analyst grounding: The backend extracts entities (year/team/player/position) from each question, attaches real parquet data as JSON context, and instructs the model to never assert a value not present in that context.

API Reference

Route	Description
GET /api/draft/{year}	All picks in a draft class with TDVS scores
GET /api/curve	Expected-EPA curve per position + Chase chart comparison
GET /api/teams	All 32 GM scorecards
GET /api/team/{abbr}	One team's full pick-by-pick history
GET /api/player/{gsis_id}	Player card with year-by-year EPA
GET /api/redraft/{year}[/{team}]	Optimized draft order + team value deltas
GET /api/draft/2026/preview	2026 draft class joined to the expected-EPA curve
POST /api/analyst	AI analyst question/answer
GET /api/methodology	Methodology page content
GET /health	Health check

Data Attribution

- nflverse / nflfastR — play-by-play and draft pick data
- nfl_data_py — Python client for nflverse data
- Stuart Chase draft value chart — traditional draft-value comparison baseline (approximated)

Methodology

1. The Problem We're Solving

NFL teams traditionally judge draft picks using career reputation, Pro Bowl selections, or static "draft value charts" built from historical trade patterns, not actual on-field production. DraftSpline instead asks: did this pick deliver value relative to what was reasonably expected from that draft slot, during the only window a team fully controls the player's economics, the four-year rookie contract?

2. What EPA Is

EPA stands for Expected Points Added. It's a play-by-play metric that estimates how much a single play changed a team's expected scoring outcome for that drive, based on down, distance, field position, and game state. A quarterback who throws a touchdown on 3rd-and-long adds a lot of EPA; a running back stopped for a loss on 1st-and-10 subtracts EPA. Summed across a season, total EPA is a context-adjusted measure of how much a player's plays actually helped (or hurt) their team score.

3. The Rookie Contract Window

Standard NFL rookie contracts run four years (with a possible fifth-year option for first-round picks, which is excluded). DraftSpline sums each player's EPA across draft year through draft year + 3, the years a team controls the player at a pre-market, below-veteran-market price. Career value beyond that window is deliberately excluded: a great Year 6 doesn't retroactively make a pick a good draft-capital decision if Years 1-4 were replacement level. To avoid unfairly penalizing injury-shortened seasons, any season with fewer than 8 games played is scaled by games_played / 17 before being summed into the rookie total.

4. How the Expected EPA Curve Is Fit

For each of the four modeled positions (QB, RB, WR, TE), every qualifying drafted player from 2012-2022 is plotted by rookie-contract EPA against overall pick number, and a smoothed curve is fit using Nadaraya-Watson kernel regression (a LOESS-style local smoother) followed by isotonic regression to enforce that expected value never increases as pick number gets worse. This produces one expected-EPA value per pick number, per position, plus a 90% confidence band that widens in later rounds, where there are fewer qualifying players per pick and the model is extrapolating more. This curve is compared against the Stuart Chase draft value chart, a widely cited update to the older Jimmy Johnson chart that NFL front offices have historically used to value trade compensation. The Chase chart is position-agnostic by design; that uniformity is itself one of DraftSpline's central findings, since real value is not position-agnostic.

5. How TDVS Is Calculated

```
TDVS = Rookie Contract EPA (adjusted) / Expected EPA at draft slot
```

A TDVS of 1.0 means a player delivered exactly what their slot predicted. Above 1.0 is a steal; below 1.0 is an underperformer.

A technical note on stabilization: rushing offense has a well-documented property in NFL analytics, average EPA per rushing play is usually negative, even for excellent running backs, because rushing is a lower expected-value play type than passing in most game states. This means the expected-EPA curve for RB (and occasionally other positions in sparse late rounds) can be at or near zero. A literal ratio is unstable near zero, so DraftSpline uses a formula that is mathematically identical to the literal ratio whenever expected EPA is comfortably positive, and transitions smoothly to a "value above expectation" form when expected EPA is small or negative. This keeps the TDVS = 1.0 "met expectation" anchor consistent everywhere without ever dividing by a near-zero number.

Players below the minimum qualifying-play threshold for their position are excluded from TDVS ranking and shown as "insufficient data":

Position	Minimum Rookie-Window Plays
QB	150 dropbacks
RB	150 carries + targets
WR	75 targets
TE	50 targets

6. Known Limitations

- OL is excluded entirely. Offensive line value requires PFF-style blocking-grade data not available in the public nflverse dataset.
- Defensive positions (DL, LB, CB, S) are not modeled in v1. Defensive EPA exists in nflfastR but requires a different curve architecture; this is explicitly scoped as a v2 feature.
- Special teams (K, P, LS) are excluded. They're modeled separately in practice and not comparable to offensive skill-position EPA.
- The 4-year rookie window is a simplification. Some players hold out, get cut before Year 4, or get extended early. Games-played normalization partially compensates, but outlier player scores can still be affected.
- Confidence intervals widen substantially in rounds 5-7. This is honest signal about real sample-size limitations, not a smoothing failure; the band is shown rather than hidden.
- EPA does not capture all player value, most notably blocking schemes, special-teams contribution, and leadership/intangibles.

7. Data Sources

• nflverse / nflfastR — play-by-play data, 2012-2025, including EPA per play.

• nfl_data_py — the Python client used to pull nflverse data.

• Stuart Chase draft value chart — a static, publicly cited chart used only as the traditional-value comparison baseline, not as an input to the TDVS model itself.

Replacement-Level Table

Expected EPA at round-break picks, by position:

Position	Pick 1	Pick 33	Pick 65	Pick 97	Pick 129	Pick 161	Pick 193+
QB	~38	~8	~1	~0	~0	~0	~0
RB	~-26	~-26	~-26	~-26	~-27	~-28	~-29
WR	~78	~65	~50	~36	~24	~14	~6
TE	~46	~40	~28	~18	~10	~5	~1

Run Locally

Backend

```
cd backend
pip install -r requirements.txt
uvicorn main:app --reload
```

Serves on http://localhost:8000. Requires backend/data/*.parquet (already committed) and a .env with GROQ_API_KEY for the AI analyst (falls back to cached responses if unset).

Frontend

```
cd frontend
npm install
npm run dev
```

Serves on http://localhost:5173, reading VITE_API_BASE_URL from .env.local (defaults to http://localhost:8000).

Pipeline (optional, outputs already committed)

```
pip install -r pipeline_requirements.txt
python 01_load_nflverse.py
python 02_compute_epa_per_player.py
python 03_compute_tdvs.py
python 04_fit_draft_curve.py
python 05_compute_team_scores.py
```

Deployment

- **Backend → Render:** root backend/, build pip install -r requirements.txt, start uvicorn main:app --host 0.0.0.0 --port $PORT. Set ALLOWED_ORIGINS and GROQ_API_KEY env vars.
- **Frontend → Vercel:** root frontend/, framework preset Vite, set VITE_API_BASE_URL to the Render URL. vercel.json handles SPA routing.
- **Keep-alive:** register the Render /health endpoint at cron-job.org on a recurring interval to avoid cold starts before a demo.